
Detecting Time Series Anomalies Like an Expert: A Multi-Agent LLM Framework with Specialized Analyzers

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Abstract

Recent studies have explored large language models for time-series anomaly detection, yet existing approaches often rely on a single general-purpose model to directly infer anomaly indices or intervals, limiting controllability, interpretability, and reliability for complex anomaly patterns. We propose SAGE (Specialized Analyzer Group for Expert-like Detection), a multi-agent framework for structured anomaly diagnosis in univariate time series. It decomposes anomaly analysis into four specialized Analyzers for point, structural, seasonal, and pattern anomalies. Each Analyzer applies family-specific numerical tools and diagnostic visualizations to generate evidence, while an evidence-grounded Detector consolidates the evidence into confidence-scored anomaly records with intervals and candidate types. A Supervisor then converts these structured records into analyst-facing diagnostic reports. SAGE further constructs synthetic in-context examples from normal-reference training segments, without using real anomalous segments or anomaly-type labels as in-context examples. Across three benchmarks, SAGE achieves the best average performance among strong ML/DL and language-model-based baselines. Ablation studies and human evaluation further show that the proposed framework improves detection reliability and the practical usefulness of diagnostic outputs.

1 Introduction

Time-series anomaly detection (TSAD) is central to applications such as industrial monitoring, financial fraud detection, medical diagnosis, and network security [13, 23]. Existing methods range from classical statistical approaches to modern deep learning models [2, 19, 30]. However, most TSAD methods focus on producing anomaly scores and provide limited support for structured explanations of anomaly types and decision evidence. Many recent deep learning approaches emphasize specific time-series characteristics, such as decomposition, Fourier analysis, or multi-resolution modeling [30, 35], whereas human experts typically diagnose anomalies by jointly considering statistics, trends, seasonality, and pattern changes. This contrast motivates a framework that separates these diagnostic perspectives into specialized analyses and integrates their evidence into a coherent

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decision. Without such decomposition, heterogeneous anomaly types, including point anomalies, change points, frequency changes, and pattern shifts, remain difficult to diagnose in an interpretable manner.

Recent advances in large language models (LLMs) and vision-language models (VLMs) offer new opportunities for interpretable TSAD, but existing LLM/VLM-based approaches still face important challenges. A single general-purpose LLM provides limited control over long numerical inputs, precise statistical operations such as change-point analysis, and simultaneous reasoning across multiple temporal perspectives [5, 18]. For example, LLMAD [18] improves LLM-based TSAD through AnoCoT reasoning and ICL with cross-retrieved real anomaly examples, but it still relies on a single LLM for final anomaly inference and is less suitable when anomaly labels are unavailable. Recent multi-agent approaches such as TSAD-Agents [31] demonstrate the potential of agentic anomaly analysis, but their decomposition remains relatively coarse and does not explicitly organize diagnostic evidence around distinct anomaly families such as point, structural, seasonal, and pattern anomalies.

We propose SAGE (Specialized Analyzer Group for Expert-like Detection), a multi-agent LLM framework for structured anomaly diagnosis in univariate time series. Instead of treating TSAD as a single anomaly-scoring problem, SAGE assigns different anomaly families to specialized Analyzers and integrates their findings into a unified diagnostic decision. The four Analyzers target point, structural, seasonal, and pattern anomalies, collecting quantitative tool outputs and diagnostic visual evidence. A Detector aggregates these outputs to produce anomaly intervals, confidence scores, and anomaly types. SAGE combines this analyzer structure with a dual-representation strategy, multimodal analysis, and synthetic ICL constructed from normal-reference training segments; binary anomaly labels are used only to exclude anomalous intervals from the reference pool, without using real anomalous segments or anomaly-type labels as in-context examples. We focus on univariate TSAD as a first step toward structured and interpretable anomaly diagnosis, leaving multivariate extensions for future work.

The main contributions of this paper are as follows:

1. We reformulate TSAD as structured diagnosis, decomposing anomaly analysis by evidence patterns and producing intervals, confidence scores, anomaly types, and diagnostic outputs.
2. SAGE introduces specialized Analyzers for point, structural, seasonal, and pattern anomalies, together with an evidence-grounded Detector that aggregates heterogeneous tool evidence.
3. SAGE builds synthetic ICL references from normal-reference training segments, without using real anomalous segments as in-context examples.
4. Experiments across benchmarks, ablations, synthetic type evaluation, human evaluation, and backbone comparisons show improved detection performance and diagnostic usefulness.

2 Related Work

Time Series Anomaly Detection. Time-series anomaly detection (TSAD) has been studied through statistical, distance- or density-based, and deep learning approaches. Statistical methods such as Z-score and ARIMA detect anomalies based on distributional assumptions or temporal dynamics, while methods such as LOF [3] and Isolation Forest [17] identify samples that deviate from normal data distributions. Deep learning methods, including LSTM-AE [19], USAD [2], and Anomaly Transformer [30], have achieved strong detection performance using reconstruction errors, latent representations, or attention-based dependency modeling. Recent methods further focus on specific time-series characteristics such as decomposition, Fourier analysis, and multi-resolution representations [30, 35]. However, these approaches primarily focus on anomaly scoring and provide limited support for structured anomaly types, decision evidence, and diagnostic explanations. SAGE extends this detection-oriented paradigm by pairing anomaly candidates with quantitative evidence and type information.

LLMs/VLMs for Time Series. Large language models (LLMs) have recently been applied to time-series analysis in various ways. PromptCast [32], LLTime [7], and Time-LLM [10] convert time-series values into text or token sequences to leverage the forecasting and reasoning capabilities of LLMs. These studies show that LLMs can be useful for interpreting and predicting time-series

patterns, but their primary focus is forecasting or general time-series reasoning rather than evidence-grounded anomaly diagnosis.

In the anomaly detection context, LLMAD [18] uses a single LLM with AnoCoT reasoning and ICL based on retrieved real anomaly examples, while SigLLM [1] converts time series into textual representations and evaluates direct prompting and forecasting-based detection pipelines. TAMA [37] converts time series into visual representations and uses few-shot multimodal prompting for anomaly detection and type-level explanation. These methods demonstrate the applicability of LLMs and VLMs to TSAD, but they vary substantially in input representation, ICL source, output format, confidence estimation, and diagnostic capability. SAGE takes a different direction by combining compressed text summaries, analyzer-generated visual evidence, synthetic ICL constructed from normal data, and evidence generation organized by anomaly family, shifting LLM-based TSAD from direct anomaly inference toward structured diagnosis.

Multi-Agent and Tool-Augmented LLM Systems. Multi-agent LLM systems have gained increasing attention for solving complex tasks. AutoGen [29], MetaGPT [8], CAMEL [14], and MALT [20] show that role decomposition and collaboration can improve problem-solving capabilities over single-agent systems. Tool-augmented frameworks such as ReAct [34] and Toolformer [25] further demonstrate that external tools can compensate for the numerical and procedural limitations of LLMs. In TSAD, TSAD-Agents [31] demonstrates the potential of agentic anomaly analysis.

However, general multi-agent systems mainly focus on generic role decomposition and collaboration mechanisms, leaving domain-specific evidence generation and aggregation protocols underexplored. SAGE addresses this gap by decomposing time-series anomaly diagnosis according to anomaly families, representing heterogeneous evidence in a shared format, and aggregating it through an explicit scoring rubric.

3 Proposed Method: SAGE

We consider the univariate TSAD setting. Given a univariate time series $X = (x_1, \dots, x_n)$, SAGE outputs a set of anomaly records:

$$\mathcal{Y} = \{(s_j, e_j, c_j, \tau_j, d_j)\}_{j=1}^m. \quad (1)$$

Here, s_j and e_j are the start and end indices, $c_j \in [0, 1]$ is the confidence score, $\tau_j \subseteq \mathcal{T}$ is a set of candidate anomaly types, and d_j denotes diagnostic evidence supporting the analyst-facing report. The set $\mathcal{T}$ denotes the anomaly-type taxonomy.

3.1 Overall Architecture

Figure 1 illustrates the overall architecture of SAGE. SAGE is a hierarchical pipeline that processes an input time series through five stages. Its key design is not to simply parallelize direct inference across multiple LLM agents, but to decompose the analysis by anomaly family and integrate quantitative and visual evidence generated by specialized Analyzers into a shared diagnostic framework. The Detector then converts the collected evidence into structured anomaly decisions, completing the pipeline as a diagnosis-oriented system.

In the Input Stage, SAGE receives a univariate time series. In the Representation Stage, the input is converted into two complementary representations. The original time series is preserved for accurate numerical computation by dedicated tools, while a compressed summary is used as a token-efficient representation for LLM prompting. In the Multi-Analyzer Stage, four specialized Analyzers examine the input time series in parallel. Each Analyzer is responsible for a specific anomaly family and collects different types of anomaly evidence using dedicated numerical tools and diagnostic visualizations. In the Detector Stage, the Analyzer outputs are aggregated to produce anomaly intervals, anomaly types, and evidence-strength-based confidence scores. The synthetic ICL module is not used for the Analyzers themselves; instead, it provides retrieval-based reference evidence at the Detector stage to support confidence scoring and anomaly type ranking. Finally, in the Diagnosis Stage, a Supervisor agent converts the structured detection results into an analyst-facing diagnosis that includes the alert level, time-series characteristics, alarm rationale, and recommended actions.

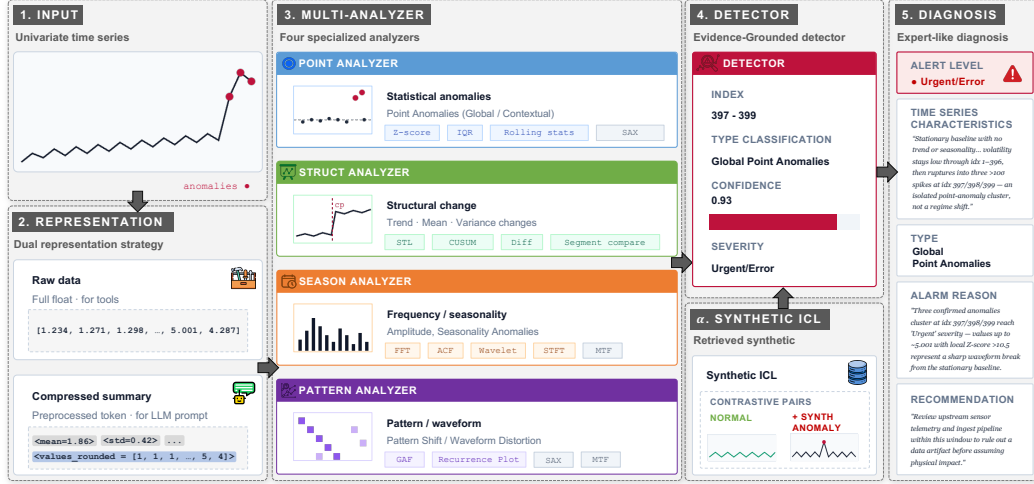


Figure 1: Overview of SAGE. The input time series is converted into dual representations and analyzed in parallel by four specialized Analyzers. Their outputs are aggregated by an evidence-grounded Detector, optionally supported by synthetic ICL, to produce anomaly predictions, which are then converted into analyst-facing diagnoses.

These stages separate roles across anomaly families and aggregate quantitative and visual evidence into diagnostic outputs, rather than relying on direct inference over raw time series.

3.2 Dual Representation Strategy

Directly providing long time series to an LLM substantially increases token cost, whereas numerical tools require access to precise original values. SAGE therefore separates each input into two representations: the original time series for tool-based computation and a compressed summary for token-efficient LLM prompting. The compressed summary contains key statistics, integer-rounded representative samples, segment-level summaries, and extrema-preserving sampled indices, with implementation details provided in Appendix C. For 400-point inputs, this reduces token usage by approximately 75%. This separation allows SAGE to reduce LLM input length without sacrificing the numerical fidelity required by statistical and signal-processing tools.

3.3 Multi-Analyzer System

SAGE groups nine anomaly types into four anomaly families and assigns one specialized Analyzer to each family. This grouping reflects the different evidence patterns required for local deviations, regime-level changes, periodic disruptions, and shape-level distortions. PointAnalyzer targets global and contextual point anomalies, primarily relying on outlier detection and rolling statistics; StructAnalyzer targets trend, mean, and variance changes through decomposition, change-point tests, and segment comparisons; SeasonAnalyzer targets amplitude and seasonality anomalies through autocorrelation, spectral analysis, and time-frequency representations; and PatternAnalyzer targets pattern shifts and waveform distortions using symbolic and recurrence-based methods. In addition to these primary tools, Analyzers may use shared auxiliary tools for visual and pattern analysis, such as GAF, MTF, recurrence plots, and symbolic representations, to provide complementary multimodal evidence. Detailed anomaly definitions and tool assignments are provided in Appendices A and H.

Let $\mathcal{A} = \{A_{pt}, A_{str}, A_{sea}, A_{pat}\}$ denote the four Analyzers. Each Analyzer maps the original series X and compressed representation $\tilde{X}$ to a family-specific analysis result:

$$\mathcal{O}_a = A_a(X, \tilde{X}), \quad a \in \{pt, str, sea, pat\}. \quad (2)$$

The outputs $\mathcal{O}_a$ may differ in their internal fields, such as outlier indices, change points, spectral summaries, recurrence patterns, or visualization-derived diagnostics. For Detector-level aggregation, these heterogeneous outputs are converted into a shared evidence representation containing candidate

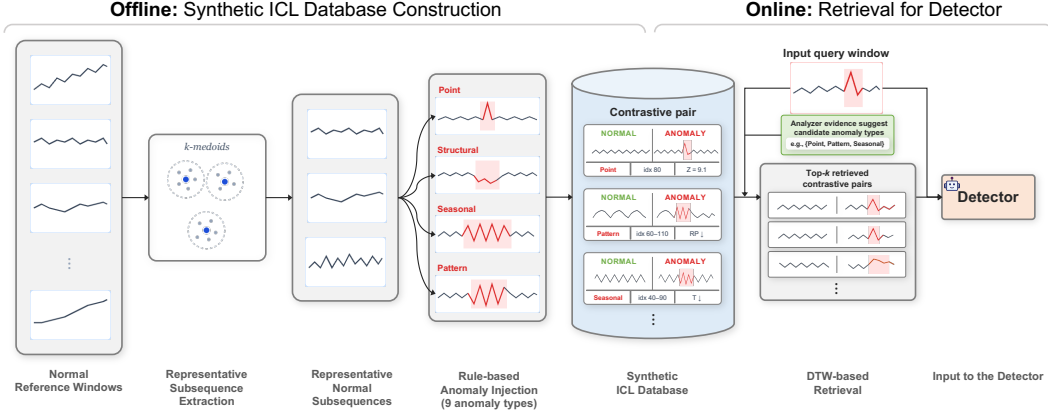


Figure 2: Synthetic ICL pipeline in SAGE. Normal subsequences are converted into synthetic contrastive references and retrieved at inference time to support Detector-level confidence scoring and anomaly type ranking.

locations or intervals, candidate anomaly types, analysis summaries, and supporting tool outputs. This representation allows the Detector to compare evidence across Analyzer families while preserving family-specific diagnostic information.

3.4 Evidence-Grounded Detector

The Detector aggregates the family-specific Analyzer outputs and analyzer-generated diagnostic visualizations to produce anomaly intervals, confidence scores, and anomaly types. Its primary role is to integrate evidence from the four Analyzers, rather than to perform standalone anomaly detection from the raw numerical time series. The diagnostic visualizations are treated as evidence artifacts that complement quantitative outputs and help verify Analyzer-supported candidates.

Let $\mathcal{O} = \{\mathcal{O}_a : a \in \{\text{pt}, \text{str}, \text{sea}, \text{pat}\}\}$ denote the collection of Analyzer outputs. The Detector maps these outputs to the anomaly record format defined in Eq. (1):

$$\mathcal{Y} = D(\mathcal{O}, \mathcal{V}, \mathcal{R}), \quad (3)$$

where $\mathcal{V}$ denotes diagnostic visualizations and $\mathcal{R}$ denotes retrieved synthetic ICL references.

For each candidate interval I , the Detector assigns a confidence score $c(I) \in [0, 1]$ through a rubric-guided scoring protocol. The LLM is instructed to rate evidence strength on a 0–100 integer scale, which is then normalized to $[0, 1]$. The rubric considers the magnitude and statistical significance of deviations, agreement across Analyzers, and the presence or absence of conflicting evidence. When evidence conflicts across Analyzers, the rubric prioritizes candidates supported by strong tool outputs and cross-Analyzer agreement, while lowering confidence for candidates whose support is weak or isolated. The rubric maps these considerations to ordinal confidence bands, producing a score that reflects evidence strength rather than a calibrated anomaly probability. Thus, confidence reflects the quality and agreement of supporting evidence rather than a simple count of agreeing Analyzers. We set temperature to 0.0 in all experiments for deterministic outputs.

3.5 Synthetic In-Context Learning

Figure 2 illustrates the synthetic ICL pipeline. SAGE constructs the ICL database from normal-reference training segments after excluding labeled anomalous intervals from the reference pool. Representative normal-reference subsequences are extracted using k -medoids clustering, and each prototype is augmented with rule-based perturbations for the nine anomaly types to form normal–anomalous contrastive references. This use of binary labels is restricted to reference-pool construction and does not provide SAGE with real anomalous demonstrations, anomaly-type supervision, or test-set information. Thus, the synthetic ICL module is designed to supply contrastive evidence rather than supervised anomaly examples. At inference time, SAGE retrieves the top- $k = 3$ most similar normal prototypes using Dynamic Time Warping (DTW) [24] with LB_Keogh pruning [12]. From

the retrieved prototypes, SAGE selects synthetic contrastive references according to the anomaly type candidates proposed by the Analyzers and provides them to the Detector to support confidence scoring and type ranking, without using test-set labels, true anomaly types, or real anomalous segments. Detailed procedures and injection rules are provided in Appendix B.

3.6 Confidence-Based Detection

Many existing LLM-based anomaly detection approaches primarily output anomaly indices or intervals [18, 31, 37], which limits users’ ability to adjust detection sensitivity or inspect supporting quantitative evidence. In contrast, SAGE assigns a continuous confidence score to each anomaly candidate, making detection a tunable diagnostic process. Users can adjust the threshold to control the trade-off between false positives and false negatives while inspecting the associated evidence and anomaly type prediction.

4 Experimental Setting

4.1 Datasets

To evaluate SAGE across different domains and time-series characteristics, we conduct experiments on three representative TSAD benchmark datasets.

Table 1: Dataset statistics.

Dataset	# Series	Avg. Length	Domain
Yahoo S5 [33]	367	~1,400	Web traffic (real + synthetic)
KPI [15]	29	~100K	IT infrastructure (AIOps)
WSD [36]	111	~17K	Web service metrics

Yahoo S5 contains real web traffic series (A1) and synthetic subsets (A2–A4), while KPI and WSD consist of IT infrastructure and web service metrics, respectively [15, 33, 36]. The three datasets differ in sequence length, anomaly patterns, and domain complexity, and all provide only binary anomaly labels without anomaly-type annotations. To evaluate type-level detection and diagnosis, we therefore construct a separate synthetic evaluation set with controlled anomaly injections. We preserve temporal order in train/test splits; the training portion is used only to extract normal patterns and construct the synthetic ICL database, not for parameter learning.

4.2 Baselines

We compare SAGE with both traditional ML/DL-based TSAD methods and recent LLM/VLM-based anomaly detection methods. ML/DL baselines include Isolation Forest [17], OCSVM [26], LOF [3], LSTM-AE [19], LSTM-VAE [22], USAD [2], OmniAnomaly [27], and Anomaly Transformer [30]. LLM/VLM baselines include SigLLM [1], LLMAD [18], TAMA [37], and TSAD-Agents [31]. For a fair framework-level comparison, all LLM/VLM baselines are evaluated with the same backbone model, Claude Sonnet 4.6. This unified-backbone protocol controls for differences in underlying model capability and allows us to isolate the effect of framework design. For each baseline, we preserve its original input representation, prompting logic, ICL strategy, output format, and post-processing procedure whenever applicable, replacing only the backbone model.

4.3 Evaluation Metrics

We evaluate anomaly detection performance using Point-F1, PA-F1 (point-adjusted F1), Affiliation-F1 [9], and Delayed-F1 ($k = 3$). Point-F1 measures point-wise detection accuracy, while PA-F1 follows the point-adjusted protocol widely used in the TSAD literature and provides a relaxed segment-level evaluation. Affiliation-F1 accounts for temporal proximity between predicted and ground-truth anomaly intervals. Delayed-F1 evaluates early detection by treating an anomaly as detected if the first prediction occurs within k time steps after the anomaly onset. We use $k = 3$ throughout the experiments.

Table 2: Main results across three benchmarks and four evaluation metrics. Best in bold, second-best underlined. "AnomalyTF", "OmniAnom", and "TSAD-Ag" denote Anomaly Transformer, OmniAnomaly, and TSAD-Agents, respectively.

Method	Yahoo S5				KPI				WSD				Average			
	Pt	PA	Aff	Del	Pt	PA	Aff	Del	Pt	PA	Aff	Del	Pt	PA	Aff	Del
iForest	0.514	0.520	0.693	0.540	0.399	<u>0.779</u>	<u>0.900</u>	0.498	0.331	0.791	0.841	0.550	0.415	0.697	0.811	0.529
OCSVM	0.505	0.522	0.675	0.533	0.399	0.758	0.896	0.539	0.350	0.876	0.916	0.741	0.418	0.719	0.829	0.604
LOF	0.461	0.474	0.673	0.478	0.066	0.138	0.689	0.107	0.104	0.262	0.728	0.282	0.210	0.291	0.697	0.289
LSTM-AE	0.547	0.557	0.830	0.574	0.278	0.696	0.844	0.266	<u>0.471</u>	0.808	0.892	0.546	0.432	0.687	<u>0.855</u>	0.462
LSTM-VAE	0.518	0.526	0.681	0.548	<u>0.422</u>	<u>0.749</u>	0.901	0.525	0.442	<u>0.920</u>	0.892	0.765	<u>0.461</u>	<u>0.732</u>	0.825	0.613
USAD	0.491	0.504	0.674	0.525	0.307	0.659	0.866	0.529	0.324	0.781	0.858	0.606	0.374	0.648	0.799	0.553
OmniAnom	0.456	0.497	0.755	0.510	0.148	0.441	0.722	0.136	0.396	0.830	0.818	0.447	0.333	0.589	0.765	0.364
AnomalyTF	0.133	0.168	0.527	0.497	0.384	0.586	0.848	<u>0.557</u>	0.423	0.600	0.912	<u>0.871</u>	0.313	0.451	0.762	0.642
SigLLM	0.539	0.550	0.864	0.581	0.117	0.273	0.689	0.046	0.287	0.590	0.696	0.195	0.314	0.471	0.750	0.274
LLMAD	<u>0.767</u>	<u>0.771</u>	<u>0.878</u>	<u>0.827</u>	0.195	0.442	0.847	0.480	0.416	0.715	0.794	0.770	0.460	0.643	0.843	<u>0.695</u>
TAMA	0.038	0.040	0.674	0.611	0.049	0.075	0.675	0.342	0.148	0.166	0.650	0.624	0.078	0.094	0.666	0.526
TSAD-Ag	0.320	0.323	0.616	0.390	0.151	0.251	0.530	0.090	0.229	0.249	0.376	0.304	0.233	0.274	0.507	0.261
SAGE	0.797	0.804	0.884	0.843	0.610	0.791	0.871	0.734	0.581	0.957	0.974	0.970	0.663	0.851	0.910	0.849

Thresholding and post-processing. Following common TSAD evaluation practice, we use post-hoc Best-F1 threshold search for methods that produce anomaly scores or confidence scores. This avoids imposing an arbitrary fixed threshold across methods with different score scales and evaluates the discriminative capacity of the produced scores. We apply the same procedure to SAGE and ML/DL baselines, and also to TAMA, which outputs discrete confidence scores. For LLM/VLM baselines whose original protocols include method-specific post-processing or threshold selection, such as LLMAD, we follow the original protocol rather than imposing an additional SAGE-specific thresholding rule. SigLLM and TSAD-Agents directly output anomaly indices or intervals without confidence scores, so the same threshold search is inapplicable. Because post-hoc threshold selection uses ground-truth labels only for evaluation, we interpret these results as offline estimates of score discriminability rather than deployable thresholding strategies. A comparison between fixed thresholds and post-hoc threshold search is provided in Appendix G.

5 Experimental Results

5.1 Main Results

Table 2 summarizes the overall performance across three benchmarks and four evaluation metrics. SAGE achieves the best average performance on all metrics, obtaining 0.663 Point-F1, 0.851 PA-F1, 0.910 Affiliation-F1, and 0.849 Delayed-F1 on average. For Point-F1, SAGE improves over the best DL baseline, LSTM-VAE, by 43.8% and over the best LLM baseline, LLMAD, by 44.1%. These results show that SAGE provides the strongest overall performance across conventional anomaly-scoring methods and recent LLM-based anomaly detection methods.

At the dataset level, SAGE consistently outperforms LLMAD in Point-F1, with especially large gains on the operational KPI and WSD datasets. These results indicate that decomposing anomaly analysis into specialized evidence generation and Detector-level aggregation improves both point-level detection and segment-level evaluation. Detailed subset-wise results for Yahoo S5 are provided in Appendix D.

5.2 Anomaly Type Analysis

Existing benchmark datasets provide only binary anomaly labels without anomaly-type annotations, and their observed anomalies appear to cover only a limited range of anomaly families. We therefore construct an additional synthetic evaluation set with injection-defined type labels to enable controlled type-wise evaluation. The type analysis in this section is computed against these synthetic labels, while author-annotated type agreement results for selected benchmark anomalies are provided in Appendix D.2. Importantly, this evaluation is conducted in a zero-shot setting without synthetic ICL, so the injected anomaly types are not exposed through retrieved contrastive examples. It therefore

Table 3: Synthetic data type evaluation and baseline comparison.

(a) Category-wise type evaluation

Category	N	Type Agr.	Det. Rec.
Point	40	88%	90%
Structural	60	93%	82%
Seasonal	24	75%	54%
Pattern	32	31%	97%
Overall	156	76%	83%

(b) SAGE vs. LLM/VLM baselines

Method	Pt-F1	PA-F1	Del-F1	Aff-F1
SigLLM	0.303	0.486	0.412	0.519
LLMAD	0.674	0.870	0.891	0.893
TAMA	0.657	0.723	0.931	0.937
TSAD-Agents	0.227	0.470	0.441	0.531
SAGE	0.723	0.879	0.896	0.910

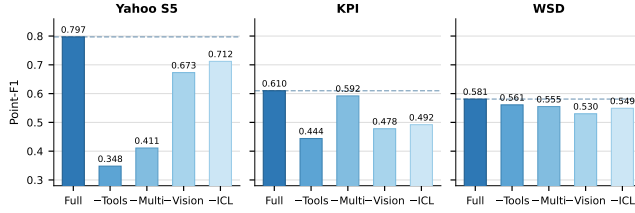


Figure 3: Point-F1 under the full SAGE system and each ablated variant across three datasets.

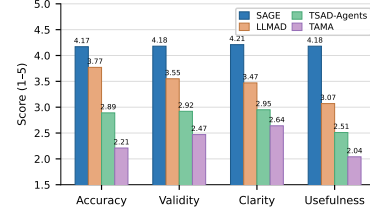


Figure 4: Average human evaluation scores across four criteria for SAGE and LLM/VLM baselines.

measures diagnostic generalization to controlled anomaly types rather than memorization of synthetic references.

Table 3a reports the category-wise type evaluation results, where Type Agr. and Det. Rec. denote Type Agreement and Detection Recall, respectively. The Structural category achieves the highest type agreement (93%), suggesting that StructAnalyzer reliably captures structural changes through change-point tests and segment-level comparisons. The Pattern category shows high detection recall (97%) but low type agreement (31%). This gap indicates that the relevant anomaly region is often localized, but distinguishing pattern shift from waveform distortion remains difficult at the final type-decision stage. The Seasonal category has the lowest detection recall (54%), indicating that frequency-domain tools are less effective for precise point-level localization, although they still provide useful diagnostic cues for seasonal changes.

Table 3b compares SAGE with LLM/VLM baselines on the synthetic evaluation set. SAGE achieves the best Point-F1 and PA-F1, indicating that specialized Analyzers and tool-based evidence aggregation improve anomaly localization and provide useful type-aware diagnostic evidence. TAMA obtains relatively high Delayed-F1 and Affiliation-F1, suggesting that visual representation-based reasoning can capture the approximate extent of anomaly intervals, even when fine-grained type diagnosis remains limited.

6 Analysis

6.1 Ablation Study

To quantify the contribution of each component, we conduct ablation studies by removing individual components from the full system. Figure 3 compares the Point-F1 of the full SAGE system and its ablated variants across datasets.

As shown in Figure 3, SAGE’s performance is driven by the interaction of multiple components, and their relative importance varies across dataset characteristics. On Yahoo, removing tool augmentation and multi-analyzer specialization substantially reduces Point-F1 from 0.797 to 0.348 and 0.411, respectively. This indicates that the gains on Yahoo are not driven by LLM prompting alone, but by quantitative tools and family-specific analysis when diverse anomaly families coexist. In contrast, on KPI and WSD, removing vision input or synthetic ICL has a relatively larger effect. In particular, removing vision input on KPI reduces Point-F1 from 0.610 to 0.478, suggesting that visual evidence helps capture subtle pattern changes in operational time series.

The effect of removing multi-analyzer specialization also depends on dataset complexity. On Yahoo, Point-F1 drops substantially from 0.797 to 0.411, whereas the drops are smaller on KPI and WSD, from 0.610 to 0.592 and from 0.581 to 0.555, respectively. This suggests that a single analyzer can still provide coarse localization when anomaly patterns are relatively simple or repetitive, but specialization becomes more important when anomaly types are diverse and mixed. Relaxed segment-level metrics can reward coarse interval coverage, so we interpret multi-analyzer specialization primarily as improving precise localization and diagnostic granularity rather than uniformly improving every metric. Synthetic ICL also improves performance across all datasets, with the largest gain on the long KPI sequences, supporting the value of richer reference patterns. Additional ablation results are provided in Appendix E.

6.2 Qualitative Evaluation (Human Evaluation)

Because quantitative metrics alone cannot fully capture diagnostic quality, we conduct a method-blind human evaluation with 10 technically experienced evaluators on 25 samples and four methods. Evaluators are shown the input series, ground-truth interval, predicted interval, and diagnosis, with method identities anonymized and presentation order randomized. The evaluators had relevant experience in time-series analysis, anomaly detection, industrial monitoring, or AI system evaluation.

We evaluate Accuracy, Validity, Clarity, and Usefulness; detailed rubrics, interface examples, per-evaluator results, and statistical tests are provided in Appendix F.

As shown in Figure 4, SAGE ranks highest across all four criteria, with an overall average score of 4.18. It obtains the highest score for Clarity (4.21), indicating that aggregating evidence from multiple specialized perspectives leads to more coherent and interpretable diagnoses. It also achieves the highest Usefulness score (4.18), reflecting its ability to go beyond anomaly decisions by providing alert levels, cause-oriented diagnostic explanations, and recommended actions.

LLMAD achieves a relatively high Accuracy score (3.77) through AnoCoT-based reasoning, but its Usefulness score remains lower at 3.07, suggesting limited support for operationally actionable information such as severity assessment or follow-up actions. In contrast, TSAD-Agents and TAMA obtain average scores below 3.0 overall. TSAD-Agents is affected by false positives from aggressive point-level detection and unstable explanations, while TAMA is limited in fine-grained diagnostic explanation despite its approximate localization ability. Under the paired Wilcoxon signed-rank protocol described in Appendix F, SAGE shows significantly higher human-evaluation scores than all baselines ($p < 0.001$). Additional qualitative case studies are provided in Appendix I.

7 Conclusion

We proposed SAGE, a multi-agent LLM framework that extends univariate TSAD from anomaly detection to structured diagnosis. SAGE decomposes anomaly analysis into point, structural, seasonal, and pattern perspectives, generates family-specific quantitative and visual evidence, and aggregates this evidence into confidence-scored anomaly records. A Supervisor then converts these structured records into analyst-facing diagnoses with anomaly types, alarm rationales, and recommended actions. By constructing synthetic ICL references from normal-reference training segments, SAGE reduces reliance on real anomalous ICL examples and anomaly-type labels.

Experiments on three TSAD benchmarks, together with synthetic type evaluation, ablation studies, and human evaluation, show that analyzer specialization, tool-grounded evidence aggregation, multimodal analysis, and synthetic ICL improve both detection performance and diagnostic usefulness. These results suggest that structured evidence generation is a promising direction for making LLM-based TSAD more controllable and interpretable.

The current framework is limited to univariate time series and requires multiple LLM calls per window, making cost and latency its main practical bottlenecks. Another limitation is that rule-based synthetic perturbations cannot cover all real-world failure modes, especially subtle system-specific anomalies whose causes are not fully observable from a single univariate signal. Future work will explore adaptive first-stage filtering, extend SAGE to multivariate time series, improve fine-grained pattern and seasonal localization, and study deployment-oriented thresholding and calibration strategies.

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Table 4: Definitions and synthetic injection rules of the nine anomaly types used in SAGE.

ID	Type Name	Family	Definition	Synthetic Injection Rule
1	Global point anomaly	Point	A point that strongly deviates from the global distribution of the series	Insert a large isolated spike or drop at a random position.
2	Contextual point anomaly	Point	A point that appears abnormal only under its local temporal context	Inject a local spike or drop within an otherwise smooth region so that it is inconsistent with nearby values.
3	Amplitude change	Seasonal	A change in the magnitude of a recurring or oscillatory pattern	Multiply a periodic segment by an amplitude scaling factor while preserving its frequency.
4	Seasonality anomaly	Seasonal	A disruption or change in periodic structure or dominant frequency	Alter the period or frequency of a periodic segment, or partially break its regular repetition.
5	Trend change	Structural	A persistent change in the slope or long-term direction of the series	Add a piecewise linear drift or slope change after a selected point.
6	Mean change point	Structural	An abrupt shift in the mean level of the series	Add a step-like level shift after a selected change point.
7	Variance change	Structural	A persistent change in the variability or dispersion of the series	Increase or decrease the local noise variance within a segment.
8	Pattern shift	Pattern	A change in the recurring temporal shape or motif structure	Replace a local motif with another motif of different shape while keeping similar duration.
9	Waveform distortion	Pattern	A deformation of the original waveform while preserving rough temporal locality	Warp, stretch, or locally deform the waveform shape without introducing a clear level shift.

Table 5: Synthetic injection parameter settings used for the paper-level anomaly taxonomy. Here n denotes sequence length and σ denotes the standard deviation of the normal prototype.

Type	Parameterization
Global point anomaly	Insert 1–3 isolated spikes or drops at random positions with magnitude 5σ .
Contextual point anomaly	Insert 1–3 local spikes using a local window of $\max(10, n/20)$ and magnitude 3 times the local standard deviation.
Amplitude change	Scale the second-half deviation from the segment mean by a factor of 2.0.
Seasonality anomaly	Modify the second-half frequency using multiplier 2.5, or flatten the second-half seasonal amplitude to 15% of its original deviation and add Gaussian noise with standard deviation 0.15σ .
Trend change	Start at $0.5n$ and add a linear trend with slope $\max(0.05\sigma, 0.05)$, reversing the direction when a strong pre-existing trend is detected.
Mean change point	Sample a change point uniformly within $[0.4n, 0.6n]$ and shift the suffix by 1.5σ .
Variance change	Sample a change point within $[0.4n, 0.6n]$ and scale suffix deviations from the segment mean by 2.0 or 2.5.
Pattern shift	Circularly shift the second half by one quarter of the detected period.
Waveform distortion	Clip the interval $[0.4n, 0.7n]$ to the local mean ± 0.5 local standard deviations.

A Anomaly Type Definitions and Injection Rules

This appendix summarizes the nine anomaly types used in SAGE and the injection rules used to construct the synthetic evaluation set and synthetic ICL database. While the main text describes SAGE at the anomaly-family level for readability, type diagnosis and synthetic data generation use a finer-grained taxonomy. Each type is defined by its computational characteristics, dominant evidence pattern, and deviation from normal temporal behavior. In synthetic data, rule-based perturbations are applied to express the defining characteristics of each type.

Table 4 summarizes the anomaly type name, parent family, definition, and representative injection rule. Although some types can be adjacent in practice, we assign the representative label based on the dominant mode of change: local isolated deviations are treated as point anomalies, persistent changes in mean, variance, or trend as structural anomalies, changes in repeated temporal structure as seasonal anomalies, and shape-level deformations as pattern anomalies.

The Point family includes local anomalies that occur at individual time steps or very short intervals. The Structural family represents persistent changes in statistical structure, such as mean, variance, or trend. The Seasonal family describes deviations in repeated temporal structures, including amplitude,

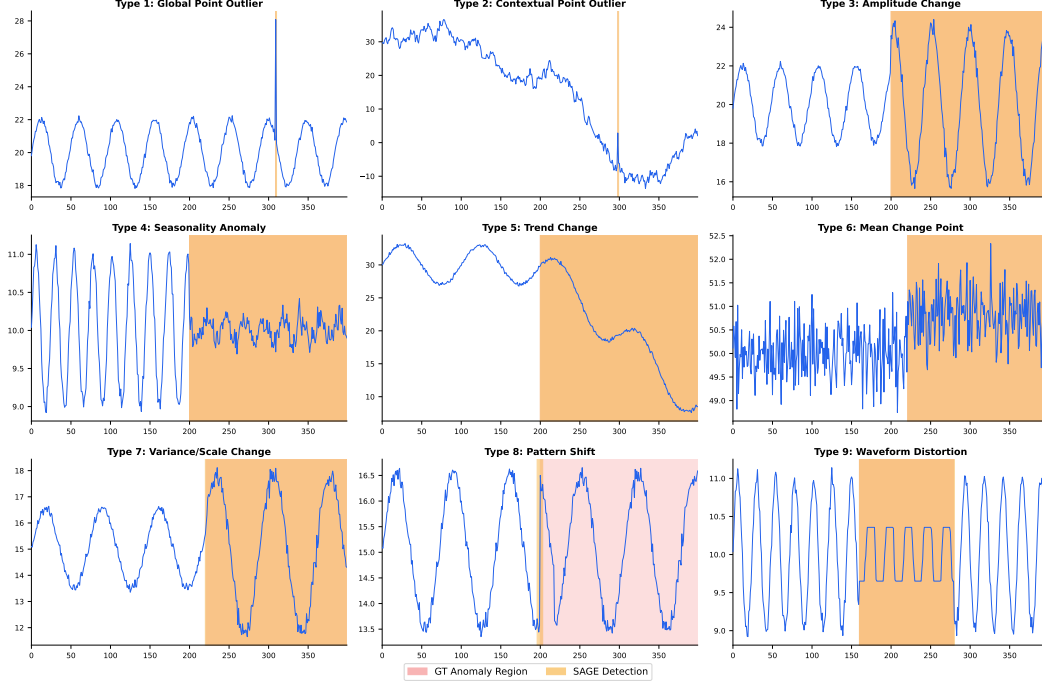


Figure 5: Representative synthetic examples of the nine anomaly types. In each panel, the red shaded region indicates the ground-truth anomaly interval, and the orange shaded region indicates the anomaly interval detected by SAGE.

periodicity, and frequency changes. The Pattern family focuses on visual or shape-level changes and waveform distortions.

To clarify boundaries between adjacent types, we apply the following rules. Abrupt changes in mean level are labeled as *mean change points*, whereas changes in long-term direction are labeled as *trend changes*. Changes in the magnitude of a repeated structure are labeled as *amplitude changes*, while disruptions or frequency changes in its periodic organization are labeled as *seasonality anomalies*. When a local motif is replaced by a different shape, it is labeled as a *pattern shift*; when the original pattern is preserved but locally deformed, it is labeled as a *waveform distortion*.

Figure 5 shows representative synthetic examples of the nine anomaly types defined in Table 4. These examples illustrate how each type appears in time series and how SAGE identifies the corresponding anomaly interval. They also serve as visual references for interpreting the type-level results on the synthetic benchmark.

B Synthetic ICL Database Construction

This appendix provides additional details on the synthetic in-context learning (ICL) database introduced in Section 3.5. The database is constructed from normal-reference training segments after excluding labeled anomalous intervals from the reference pool, and stores synthetic contrastive references for the anomaly types used by SAGE. At inference time, retrieved normal prototypes are used to select synthetic references according to the anomaly type candidates proposed by the Analyzers.

B.1 Representative Subsequence Extraction

We first extract representative subsequences that capture normal patterns in the training portion. Each training series is divided into fixed-length subsequences, and subsequences overlapping labeled anomalous intervals are excluded from the reference pool. k -medoids clustering [11] is then applied to the remaining normal-reference subsequences. The number of medoids is selected by the silhouette

score within $k \in [2, \min(20, \lfloor N/2 \rfloor)]$. If the number of segments is at most 12, we keep all segments without clustering. We use k -medoids because it preserves actual observed subsequences as prototypes, rather than using averaged centroids. The selected medoids serve as normal prototypes for subsequent anomaly injection and retrieval.

B.2 Rule-based Anomaly Injection

For each normal prototype, the TS Generator injects rule-based perturbations according to the anomaly taxonomy in Appendix A. Point anomalies are generated by inserting local spikes or drops; structural anomalies introduce regime-level changes such as mean shifts, variance changes, or trend changes; seasonal anomalies modify amplitude or periodic structure; and pattern anomalies distort motifs or waveform shapes. The injection location and magnitude are sampled within bounded ranges so that the generated examples preserve the base normal pattern while expressing the intended anomaly type.

B.3 Contrastive Pair Construction

After anomaly injection, each normal prototype is paired with its anomalous version to form a normal-anomalous contrastive pair. Because the two sequences share the same base pattern, the pair helps the Detector compare how tool-based evidence changes between normal and anomalous cases. For each normal prototype, we generate exactly one synthetic example for each of the nine anomaly types, so the database contains balanced type-complete reference sets at the prototype level. Each contrastive pair stores the normal subsequence, the injected anomalous subsequence, the anomaly type label, and a summary of tool-based evidence such as statistical changes, change-point responses, spectral changes, autocorrelation changes, and segment comparisons. Thus, the database acts as a structured reference library containing both sequence-level contrasts and evidence traces.

B.4 DTW-based Retrieval

At inference time, SAGE retrieves references by comparing the input series with representative normal prototypes in the synthetic ICL database. Similarity is computed using Dynamic Time Warping (DTW), which is robust to local temporal misalignment, and LB_Keogh pruning is used to reduce unnecessary comparisons. We select the top- $k = 3$ most similar normal prototypes. From the contrastive references associated with these prototypes, SAGE then selects references according to the anomaly type candidates proposed by the Analyzers and provides them to the Detector. Thus, Analyzer-predicted candidate types are used for reference selection after prototype retrieval, but not for computing the DTW similarity itself. Retrieval and reference selection do not use test-set ground truth, true anomaly types, or real anomalous segments.

B.5 Role of Synthetic ICL References

The synthetic ICL database allows SAGE to use in-context examples without real anomalous ICL examples or anomaly-type labels. Unlike retrieval-based ICL methods that depend on real anomalous intervals from other files, SAGE uses synthetic contrastive references generated from normal-reference prototypes. This design reduces dependence on real anomalous ICL examples while providing the Detector with contrastive evidence for confidence scoring and type diagnosis.

C Implementation Details

SAGE is implemented as a LangGraph-based workflow in which the four Analyzers are executed in parallel. Unless otherwise specified, we use a window size of 400, a stride of 400, temperature 0.0, and top- $k = 3$ retrieval for synthetic ICL. Synthetic ICL retrieval first selects normal prototypes using DTW, and then selects contrastive references according to the anomaly type candidates proposed by the Analyzers. We set the temperature to 0.0 in all SAGE experiments to ensure deterministic and reproducible outputs, since stochastic sampling can introduce unnecessary variance in point-level confidence scores and anomaly index predictions.

C.1 Compact Summary Construction

The compact LLM input is constructed from the original time series without modifying the values used by numerical tools. For each window, SAGE records length, range, mean, standard deviation, and segment-level statistics. If the full sequence exceeds the prompt budget, it samples representative indices at a uniform stride determined by the estimated token budget and appends the global minimum and maximum indices if they are not already selected. The sampled values are rounded to integers for prompt compactness, while all tool computations use the original floating-point values. In the default setting, the workflow uses a 400-point input window and a compact prompt budget of 200–500 estimated tokens depending on the agent role.

C.2 Tool Parameter Settings

Tool parameters are fixed across datasets unless they are inferred from the window length or an automatically detected period; they are not tuned per dataset. Outlier detection uses Z-score threshold 3.0 and IQR multiplier 1.5. Rolling statistics use multi-scale local windows and flag contextual candidates with local Z-score threshold 2.5. Decomposition auto-detects the period by FFT when unspecified and uses moving-average or STL-style decomposition depending on the tool call. SAX and MTF use alphabet/bin size 10, recurrence plots use neighborhood percentile 0.1, and STFT uses window size 64 with hop size 32. Regime expansion uses 50-point verification chunks, a 100-point reference window, and $p < 0.05$ for mean or variance difference tests.

For LLM/VLM baselines, we preserve the core framework and inference procedure of each original method as much as possible, while using the same backbone model, Claude Sonnet 4.6, to prevent differences in model capability from dominating the comparison. Specifically, we follow each baseline’s agent structure, prompting logic, modality usage, and output format whenever applicable, replacing only the underlying model with the common backbone. We do not perform additional prompt re-tuning, except for minimal input-output formatting required by the backbone replacement. For LLM/VLM baselines, including LLMAD, TSAD-Agents, SigLLM, and TAMA, we follow the hyperparameters reported in the original papers as closely as possible.

The main model used for evaluation is Claude Sonnet 4.6. We additionally compare SAGE across GPT-5.1, Claude Haiku 4.5, Qwen3.5-27B, and Qwen3.5-9B. API-based experiments with Claude Sonnet 4.6, Claude Haiku 4.5, and GPT-5.1 are executed through hosted model APIs and do not use local GPUs for model inference. Open-source models are served using vLLM: Qwen3.5-27B is run in FP16 on four NVIDIA A100 80GB GPUs with tensor parallel size 4, and Qwen3.5-9B is run in BF16 on one NVIDIA A100 80GB GPU. The software environment uses Python 3.10.12, PyTorch 2.10.0 with CUDA 12.8, vLLM 0.17.1, LangChain 0.3.27, LangGraph 0.6.7, Anthropic SDK 0.77.0, and OpenAI SDK 2.16.0. Based on Yahoo S5 experiment logs, the average per-window runtime was approximately 80 seconds for Claude Sonnet 4.6, 34 seconds for GPT-5.1, 481 seconds for Qwen3.5-27B, and 44 seconds for Qwen3.5-9B. In a representative Sonnet 4.6 run on Yahoo A1, each 400-point window used approximately 22.4K input tokens and 4.9K output tokens.

Confidence-based thresholding is applied only to methods that output continuous anomaly scores or confidence scores. We apply post-hoc Best-F1 threshold search to SAGE and score-based ML/DL baselines, and also to TAMA, which outputs discrete confidence scores. For LLM/VLM baselines, we preserve each method’s original post-processing and threshold-selection protocol whenever applicable; for example, LLMAD is evaluated using its original interval-length-based post-processing rather than an additional confidence threshold. SigLLM and TSAD-Agents directly generate anomaly indices or intervals without continuous confidence scores, so the same threshold search cannot be applied. Thus, post-hoc thresholding is treated as a common offline TSAD evaluation protocol for score-based outputs, not as a deployment-time thresholding strategy.

D Additional Experimental Results

This appendix reports experimental results that are not included in the main text, including subset-wise results on Yahoo S5, detailed dataset-level evaluation metrics, additional zero-shot comparisons, and model-wise results. These analyses complement the main findings and provide a finer-grained view of SAGE’s performance characteristics.

Table 6: Subset-wise results on Yahoo S5. Best results are shown in **bold** and second-best results are underlined.

Method	A1				A2				A3				A4			
	Pt	PA	Aff	Del	Pt	PA	Aff	Del	Pt	PA	Aff	Del	Pt	PA	Aff	Del
SigLLM	0.372	0.391	0.781	0.441	0.750	0.769	0.892	0.812	0.492	0.514	0.835	0.528	0.544	0.527	0.848	0.543
LLMAD	0.520	0.541	0.788	0.702	0.890	0.890	0.938	0.915	0.857	0.857	0.924	0.886	0.697	0.697	0.823	0.750
TAMA	0.021	0.025	0.611	0.564	0.041	0.043	0.673	0.602	0.032	0.037	0.651	0.598	0.057	0.055	0.762	0.681
TSAD-Agents	0.160	0.181	0.472	0.201	0.481	0.503	0.702	0.522	0.294	0.301	0.588	0.315	0.312	0.307	0.701	0.524
SAGE	0.753	0.787	0.918	0.911	0.861	0.868	0.944	0.876	0.839	0.839	0.895	0.862	0.715	0.715	0.790	0.751

Table 7: Type agreement on author-annotated anomaly labels.

(a) Overall type agreement					(b) Per-type agreement		
Model	Overall	Yahoo	KPI	WSD	Anomaly Type	N	Agreement
Sonnet 4.6	98.0%	97.7%	98.5%	100.0%	Global point anomaly	153	99.3%
GPT-5.1	84.7%	80.8%	91.8%	100.0%	Contextual point anomaly	98	95.9%
					Mean change point	39	100.0%
					Pattern shift	10	100.0%
					Overall	300	98.0%

D.1 Yahoo Subset-wise Results

Yahoo S5 contains both real data (A1) and synthetic data (A2–A4), so aggregate performance may obscure subset-specific behavior. We therefore report detailed results for each subset using Point-F1, PA-F1, Affiliation-F1, and Delayed-F1. In particular, A1 contains anomalies collected from real service environments, making it important for evaluating performance under realistic operational conditions.

As shown in Table 6, SAGE and LLMAD exhibit different performance patterns across Yahoo S5 subsets. On A1, which consists of real service data, SAGE outperforms LLMAD across all evaluation metrics. In particular, SAGE achieves a Point-F1 of 0.753 compared with 0.520 for LLMAD, and also obtains Affiliation-F1 and Delayed-F1 scores of 0.918 and 0.911, respectively. These results indicate that SAGE better handles the heterogeneous anomaly patterns observed in real service data.

The synthetic subsets show more nuanced behavior. On A2 and A3, LLMAD achieves slightly higher Point-F1, PA-F1, and Delayed-F1, suggesting that retrieval-based ICL and single-LLM reasoning can work effectively for regular and controlled synthetic anomalies. However, SAGE again outperforms LLMAD on A4 in Point-F1, PA-F1, and Delayed-F1, and also achieves higher Affiliation-F1 on A2. Overall, while LLMAD remains competitive on some simpler synthetic subsets, SAGE performs better on real data and more complex synthetic subsets.

D.2 Benchmark Interval Type Agreement

Since existing benchmark datasets do not provide anomaly-type labels, we manually annotate a subset of benchmark anomaly intervals. Specifically, we randomly select 100 anomalous files in a balanced manner across datasets, and one annotator assigns primary and secondary anomaly types to 626 anomaly intervals based on visual inspection and statistical analysis. Following the author-annotation setting used in LLMAD, this analysis is intended as a qualitative supplement rather than a fully supervised type benchmark. The results in Table 7 are computed on the 300 anomaly intervals that belong to the test split, and type agreement is evaluated against the primary anomaly type.

Table 7a reports the overall type agreement with the author-annotated anomaly labels, and Table 7b provides the per-type agreement results. Overall, SAGE with Sonnet 4.6 achieves high type agreement for the anomaly types represented in this subset. This suggests that the structured Analyzer evidence can support anomaly type diagnosis in addition to detection. However, this result should be interpreted as an auxiliary qualitative analysis based on a single annotator, intended to complement rather than replace the controlled synthetic type evaluation.

Table 8: Comparison of ICL and zero-shot settings for SAGE and LLMAD.

Dataset	Method	Pt	PA	Aff	Del
Yahoo	LLMAD (ICL)	0.767	0.771	0.878	0.827
	LLMAD (Zero-shot)	0.763	0.768	0.903	0.817
	SAGE (ICL)	0.797	0.804	0.884	0.843
	SAGE (Zero-shot)	0.712	0.720	0.875	0.813
KPI	LLMAD (ICL)	0.195	0.442	0.847	0.480
	LLMAD (Zero-shot)	0.109	0.285	0.778	0.441
	SAGE (ICL)	0.610	0.791	0.871	0.734
	SAGE (Zero-shot)	0.492	0.649	0.785	0.676
WSD	LLMAD (ICL)	0.416	0.715	0.794	0.770
	LLMAD (Zero-shot)	0.244	0.416	0.780	0.674
	SAGE (ICL)	0.581	0.957	0.974	0.970
	SAGE (Zero-shot)	0.549	0.944	0.956	0.952

Table 9: Backbone-wise results of SAGE across three benchmarks. Best results are shown in **bold**.

Model	Yahoo				KPI				WSD			
	Pt	PA	Aff	Del	Pt	PA	Aff	Del	Pt	PA	Aff	Del
Sonnet 4.6	0.797	0.804	0.884	0.843	0.610	0.791	0.871	0.734	0.581	0.957	0.974	0.970
GPT-5.1	0.509	0.522	0.692	0.612	0.391	0.548	0.771	0.568	0.546	0.940	0.972	0.970
Qwen3.5-27B	0.545	0.555	0.709	0.647	0.275	0.418	0.758	0.541	0.465	0.835	0.873	0.904
Haiku 4.5	0.429	0.436	0.735	0.700	0.223	0.357	0.716	0.554	0.465	0.761	0.796	0.889
Qwen3.5-9B	0.076	0.077	0.762	0.495	0.039	0.085	0.794	0.173	0.202	0.265	0.875	0.508

D.3 Zero-shot Comparison with LLMAD

To isolate whether SAGE’s performance gains come from its core architecture rather than external in-context examples, we compare zero-shot LLMAD without cross-retrieved ICL and zero-shot SAGE without synthetic ICL. By removing retrieval-based ICL from both methods, this comparison contrasts single-LLM anomaly reasoning with SAGE’s multi-agent architecture based on specialized Analyzers and tool-supported evidence aggregation.

As shown in Table 8, the effect of ICL in LLMAD varies substantially across datasets. On Yahoo, removing ICL has only a minor impact, with Point-F1 decreasing from 0.767 to 0.763. In contrast, KPI and WSD show much larger performance drops. On KPI, Point-F1 decreases from 0.195 to 0.109, and on WSD from 0.416 to 0.244, corresponding to relative drops of 44.1% and 41.3%, respectively. This suggests that LLMAD depends more strongly on external ICL examples in complex and heterogeneous operational time-series settings.

SAGE also shows performance degradation in the zero-shot setting, but the drops are much smaller on KPI and WSD, at 19.3% and 5.5%, respectively. More importantly, even without synthetic ICL, SAGE achieves Point-F1 scores of 0.492 on KPI and 0.549 on WSD, outperforming not only zero-shot LLMAD but also LLMAD with ICL enabled. This suggests that SAGE’s gains are not solely attributable to the presence of in-context examples, but are also supported by its core architecture based on specialized Analyzers, tool-supported evidence, and evidence aggregation.

On Yahoo, SAGE also decreases from 0.797 to 0.712 when synthetic ICL is removed, indicating that synthetic ICL provides useful support across both real and synthetic patterns. Nevertheless, the results on KPI and WSD suggest that, in operational time-series settings, architecture-level specialization plays a more fundamental role than retrieval-based ICL alone.

D.4 Backbone-wise Results

SAGE can be executed with different LLM backbones under the same multi-analyzer framework. To examine how backbone choice affects performance, we compare Claude Sonnet 4.6, GPT-5.1, Claude Haiku 4.5, Qwen3.5-27B, and Qwen3.5-9B. This analysis allows us to study the trade-off among reasoning capability, structured output stability, and cost-performance efficiency.

Table 10: Detailed ablation results. Each row removes one component from the full SAGE system. Best results are shown in **bold**.

Benchmark	Setting	Metrics			
		Pt	PA	Aff	Del
Yahoo Overall	Full	0.797	0.804	0.884	0.843
	- Multi-analyzer	0.411	0.429	0.816	0.582
	- Tools	0.348	0.367	0.747	0.545
	- ICL	0.712	0.720	0.875	0.813
	- Vision	0.673	0.681	0.838	0.770
KPI	Full	0.610	0.791	0.871	0.734
	- Multi-analyzer	0.592	0.796	0.911	0.768
	- Tools	0.444	0.685	0.816	0.706
	- ICL	0.492	0.649	0.785	0.676
	- Vision	0.478	0.617	0.799	0.691
WSD	Full	0.581	0.957	0.974	0.970
	- Multi-analyzer	0.555	0.989	0.978	0.981
	- Tools	0.561	0.962	0.973	0.970
	- ICL	0.549	0.944	0.956	0.952
	- Vision	0.530	0.927	0.959	0.959

As shown in Table 9, Sonnet 4.6 provides the most consistent performance across the three datasets and achieves the best results on most metrics. In particular, it performs best on all KPI metrics and achieves the highest Point-F1, PA-F1, and Affiliation-F1 on WSD. GPT-5.1 matches Sonnet 4.6 on WSD Delayed-F1 with a score of 0.970, but shows lower overall performance on Yahoo and KPI.

Qwen3.5-27B and Haiku 4.5 show intermediate performance, with more noticeable degradation on the operational datasets. Qwen3.5-9B performs poorly on Point-F1 and PA-F1, suggesting that it is less suitable for the current SAGE pipeline. Its Affiliation-F1 is relatively less degraded, indicating that it can sometimes preserve coarse temporal proximity, but lacks the precision and structured reasoning needed for accurate point-level detection.

D.5 Protocol for Synthetic Type Evaluation

This section describes the evaluation protocol used for the category-wise synthetic type evaluation in Table 3a. The ground-truth anomaly type of each synthetic sample is automatically assigned according to the anomaly injection rule defined in Appendix A. Thus, each synthetic sample has one target anomaly type at generation time, and type agreement is computed by comparing this injected type with the representative anomaly type predicted by SAGE.

Detection recall is defined as the fraction of samples for which the Analyzer assigned to the injected anomaly family produces at least one evidence interval overlapping the ground-truth anomaly interval. Type agreement is computed among successfully detected samples as the fraction for which the predicted representative anomaly type matches the ground-truth injection type. Since SAGE may predict multiple anomaly types for a single sample, we use the type with the highest confidence as the representative type.

E Detailed Ablation Results

This appendix provides detailed ablation results summarized in the main text. Each experiment is conducted by removing a specific component from the full system, and we report all four evaluation metrics for each dataset.

Table 10 shows detailed performance changes when each component is removed. Overall, removing either tool augmentation or multi-analyzer specialization has the largest impact on Yahoo, while removing vision input or synthetic ICL has relatively stronger effects on KPI and WSD. On KPI and WSD, some segment-level metrics, including PA-F1, Affiliation-F1, and Delayed-F1, remain similar or even slightly improve when the multi-analyzer module is removed. This suggests that coarse localization can still be preserved when anomaly patterns are relatively simple or repetitive. These relaxed metrics can favor broader interval coverage, whereas Point-F1 is more sensitive to precise localization and false positives. Nevertheless, the full system generally maintains the most stable Point-F1 performance and provides the most detailed type-specific evidence for diagnosis.

Table 11: Composition of samples used in the human evaluation.

Dataset	# Samples	Sampling Principle
Yahoo S5	8	Randomly sampled across A1–A4
KPI	4	Randomly sampled anomalous files
WSD	4	Randomly sampled anomalous files
Synthetic	9	One random sample per anomaly type

Table 12: Likert-scale criteria and anchor descriptions used in the human evaluation.

Criterion	Score	Anchor Description
Accuracy	1	Major anomalies are missed, or many normal regions are falsely identified as anomalous.
	2	Important anomalies are missed, or substantial false positives are present.
	3	Major anomalies are detected, but some misses or false positives remain.
	4	Most anomalies are correctly detected, with only minor misses or false positives.
	5	All anomalies are correctly detected with no false positives.
Validity	1	The diagnosed anomaly type, severity, or cause is entirely inconsistent with the observed data.
	2	The diagnosis is largely inaccurate or weakly supported by the data.
	3	Some parts of the diagnosis are plausible, but others are not well supported by the data.
	4	Most diagnostic conclusions are plausible and grounded in the observed evidence.
	5	All diagnostic conclusions are well supported by the data and closely align with expert judgment.
Clarity	1	The explanation is absent, unintelligible, or unrelated to the data.
	2	The explanation is vague and contains substantial logical gaps, making it difficult to follow.
	3	The overall logic is understandable, but some parts are weakly justified or difficult to interpret.
	4	The explanation is mostly clear, logically structured, and supported by relevant evidence.
	5	The explanation is highly clear, coherent, and easy to understand, with strong evidence support throughout.
Usefulness	1	The analysis is not useful for practical decision-making.
	2	Some information is provided, but practical use remains limited.
	3	The analysis provides partially useful information, but additional interpretation is needed before action can be taken.
	4	The analysis is directly useful for practical decision-making.
	5	The analysis provides concrete and actionable insights that can support immediate operational response.

F Human Evaluation Details

This appendix provides additional details on the human evaluation protocol, sample composition, evaluator instructions, interface, Likert-scale anchor rubric, per-evaluator results, and statistical tests. The predicted intervals shown to evaluators were taken from the same method outputs used in the quantitative evaluation protocol; no additional thresholding or sample-specific adjustment was performed for the human-evaluation interface.

F.1 Sample Composition

Evaluation samples were selected from Yahoo S5, KPI, WSD, and the synthetic evaluation set. Samples from the benchmark datasets were randomly drawn within each dataset. For the synthetic data, we included one sample per anomaly type to ensure balanced coverage of the type taxonomy, and each sample was randomly selected within its corresponding type.

F.2 Evaluation Rubric

The evaluation is conducted on a 5-point Likert scale across four criteria: Accuracy, Validity, Clarity, and Usefulness. Table 12 summarizes the anchor descriptions for each evaluation criterion.

F.3 Evaluator Instructions and Interface

Evaluators were instructed to assess method outputs for each sample using the four criteria in Table 12: Accuracy, Validity, Clarity, and Usefulness. They were shown the input time series, ground-truth anomaly interval, predicted interval, and diagnostic explanation for each method, with method identities anonymized and presentation order randomized. An example of the evaluation interface is shown in Figure 6. Evaluators participated voluntarily and were compensated with a small honorarium consistent with local institutional practice.

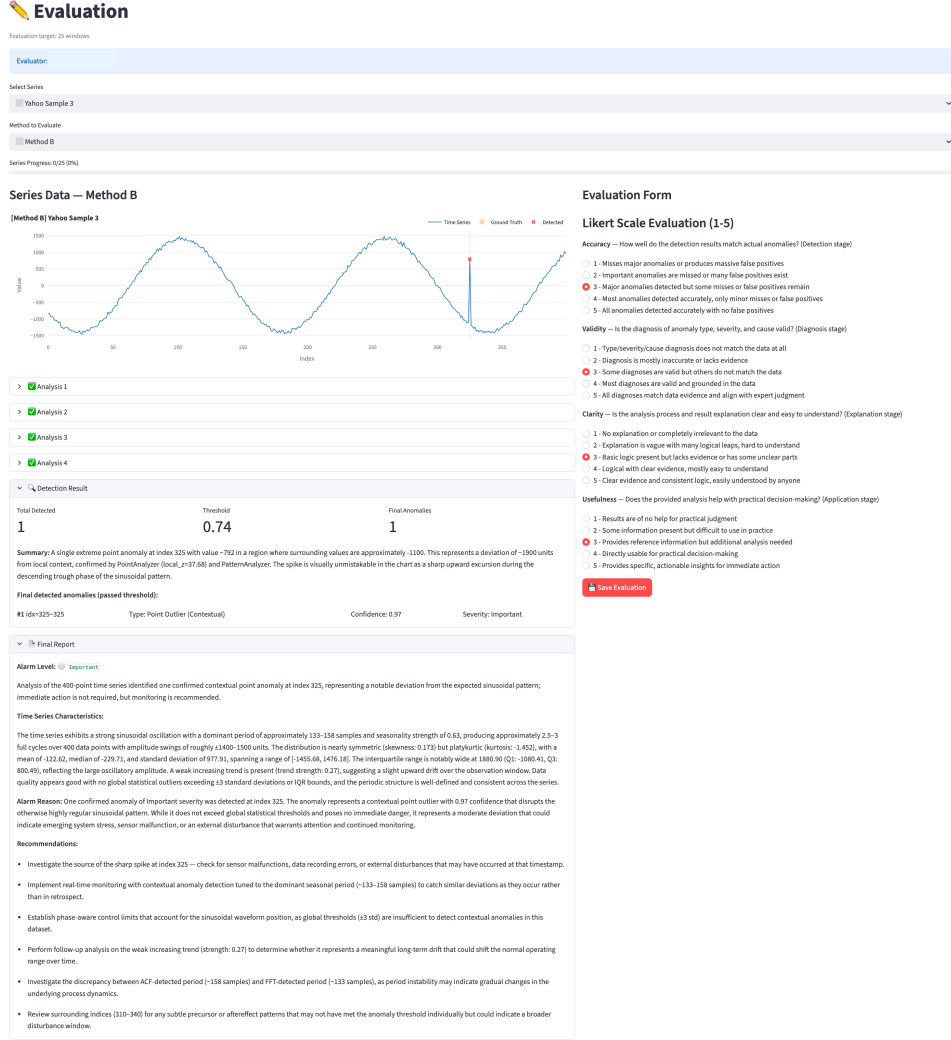


Figure 6: Example interface used for the method-blind human evaluation. Evaluators were shown the input time series, ground-truth interval, predicted interval, and diagnostic explanation for an anonymized method, and assigned Likert-scale scores for Accuracy, Validity, Clarity, and Usefulness.

F.4 Evaluator-Level Robustness Analysis

To examine whether the aggregate human evaluation results are driven by a small subset of evaluators, we provide an anonymized evaluator-level robustness analysis in Table 13. Evaluators are represented only by anonymous indices, and no personally identifiable information or evaluator identities are reported. Each cell shows the average score over 25 samples for each method and criterion.

SAGE obtains the highest average score for each criterion and the highest overall average. The advantage is consistently observed across the anonymized evaluator-level aggregates, suggesting that the main human evaluation results are not driven by a small subset of raters.

F.5 Statistical Testing

Table 14 reports the Wilcoxon signed-rank test results for pairwise comparisons between SAGE and each baseline. We conduct the test on evaluator-sample paired aggregate scores, resulting in $n = 250$ paired observations ($= 10$ evaluators $\times 25$ samples). Because evaluator-sample scores are not fully independent, we report per-expert aggregate results in Table 13 as a complementary robustness check.

Table 13: Anonymized evaluator-level robustness analysis for the human evaluation scores (5-point Likert). Evaluators are represented by anonymous indices, and each cell shows the average score over 25 samples. The bottom two rows report the mean and standard deviation across evaluators. No evaluator identities or personally identifiable information are disclosed.

Expert	SAGE (Ours)					LLMAD					TSAD-Agents					TAMA				
	Acc	Val	Cla	Use	Avg	Acc	Val	Cla	Use	Avg	Acc	Val	Cla	Use	Avg	Acc	Val	Cla	Use	Avg
E1	4.20	3.76	3.68	3.52	3.79	3.92	3.48	3.56	3.12	3.52	3.12	2.88	2.96	2.56	2.88	2.48	2.36	2.48	1.92	2.31
E2	4.16	4.08	4.16	4.28	4.17	3.80	3.20	3.72	2.84	3.39	2.48	2.96	3.36	2.56	2.84	2.08	2.76	2.92	2.08	2.46
E3	3.96	4.20	3.88	3.96	4.00	3.68	3.80	3.68	3.32	3.62	3.04	3.08	3.44	2.88	3.11	2.12	2.36	2.40	2.20	2.27
E4	4.16	4.16	3.96	3.96	4.06	4.00	4.12	4.12	4.00	4.06	2.52	2.72	2.48	2.56	2.57	1.88	2.04	2.08	1.88	1.97
E5	4.24	4.32	4.44	4.36	4.34	3.80	3.76	4.00	3.40	3.74	3.16	3.32	3.28	2.76	3.13	2.32	3.08	3.04	2.48	2.73
E6	3.56	3.76	3.76	3.48	3.64	3.20	3.00	2.68	2.52	2.85	2.80	2.64	2.40	2.08	2.48	1.88	2.44	2.24	1.76	2.08
E7	4.52	4.80	4.84	4.76	4.73	4.04	4.12	3.92	3.84	3.98	3.16	3.48	3.64	3.08	3.34	2.28	2.48	3.20	2.12	2.52
E8	4.32	4.08	4.44	4.32	4.29	3.76	3.20	3.28	2.76	3.25	3.12	3.12	3.00	2.52	2.94	2.44	2.60	2.68	2.04	2.44
E9	4.32	4.52	4.76	4.76	4.59	4.00	3.48	2.40	1.88	2.94	2.84	2.76	2.04	1.76	2.35	2.12	2.32	2.20	1.60	2.06
E10	4.24	4.16	4.20	4.36	4.24	3.52	3.36	3.36	3.00	3.31	2.64	2.24	2.92	2.36	2.54	2.52	2.24	3.20	2.36	2.58
Mean	4.17	4.18	4.21	4.18	4.18	3.77	3.55	3.47	3.07	3.47	2.89	2.92	2.95	2.51	2.82	2.21	2.47	2.64	2.04	2.34
Std	0.25	0.30	0.38	0.41	0.31	0.25	0.37	0.53	0.59	0.40	0.25	0.34	0.48	0.35	0.32	0.23	0.30	0.41	0.26	0.24

Table 14: Wilcoxon signed-rank test results for pairwise comparisons against SAGE.

Comparison	n	Test Statistic	p -value
SAGE vs. LLMAD	250	3750	< 0.001
SAGE vs. TSAD-Agents	250	790	< 0.001
SAGE vs. TAMA	250	107	< 0.001

G Threshold Strategy Analysis

Table 15 compares fixed thresholds and post-hoc Best-F1 threshold search across all evaluation metrics. Overall, fixed thresholds lead to substantial performance degradation on all datasets, with the largest impact observed on KPI. For example, Point-F1 on KPI is only 0.056 with Fixed 50 and 0.073 with Fixed 80, but increases to 0.610 with Best-F1 threshold search. The same trend is observed for PA-F1, Affiliation-F1, and Delayed-F1, indicating substantial variation in confidence distributions across files and datasets.

Yahoo also shows a large gap between fixed thresholds and Best-F1. Fixed 80 improves Point-F1 and PA-F1 over Fixed 50, but decreases Affiliation-F1, suggesting that a higher global cutoff can reduce some false positives while also removing temporally useful detections. A similar pattern is observed on WSD, where Fixed 80 outperforms Fixed 50 but still remains far below Best-F1.

These results show that a single global cutoff is insufficient for stable detection in heterogeneous time-series environments. Although the Detector is prompted with a 0–100 evidence-strength rubric, the resulting score should be interpreted as an ordinal evidence-strength score rather than a probability-calibrated output, and its distribution can shift across datasets and windows. In this context, post-hoc Best-F1 threshold search should be interpreted as an evaluation-only protocol that estimates the achievable performance of confidence-based detection, rather than as a deployment-time thresholding strategy. Deployment would require calibration or validation-set threshold selection. In practice, the value of SAGE lies not only in its confidence score, but also in its structured evidence, anomaly type information, and natural-language explanations, which can help practitioners choose operating thresholds using domain knowledge.

H Tools Used by Each Analyzer

This section summarizes the main numerical and visualization tools used by the four Analyzers. These tools are grouped into five categories: statistical outlier detection, structural change testing, frequency analysis, symbolic and pattern analysis, and visual representation. Rather than relying on a single general-purpose model, SAGE performs anomaly diagnosis based on the quantitative and visual evidence produced by these tools.

Table 15: Comparison of fixed thresholds and post-hoc Best-F1 threshold search across all evaluation metrics. Best results are shown in **bold**.

Strategy	Yahoo				KPI				WSD			
	Pt	PA	Aff	Del	Pt	PA	Aff	Del	Pt	PA	Aff	Del
Fixed 50	0.256	0.259	0.772	0.655	0.056	0.082	0.685	0.231	0.122	0.163	0.702	0.550
Fixed 80	0.537	0.541	0.735	0.677	0.073	0.170	0.735	0.413	0.258	0.395	0.696	0.680
Best-F1	0.797	0.804	0.884	0.843	0.610	0.791	0.871	0.734	0.581	0.957	0.974	0.970

Table 16: Main tools used by the four Analyzers. P denotes a primary tool for the corresponding Analyzer, and S denotes a shared auxiliary tool.

Analyzer	Tool	Function	Role
Point	statistics	Mean, standard deviation, skewness, kurtosis	P
	outliers	Global Z-score and IQR-based outlier detection	P
	rolling_statistics	Multi-scale rolling statistics	P
	symbolic_repr	SAX-based symbolic representation	S
	recurrence_plot	Recurrence-plot-based structural analysis	S
Struct	decomposition	STL-based trend/seasonal/residual decomposition	P
	diff	First- and second-order differences	P
	change_point	CUSUM-based change-point detection	P
	segment_comparison	Segment-level statistical comparison	P
	GAF	Gramian Angular Field (VLM)	S
Season	recurrence_plot	Recurrence-plot-based structural analysis	S
	auto_correlation	Split-half ACF comparison	P
	fourier_transform	FFT-based frequency component extraction	P
	wavelet_transform	Wavelet-based time-frequency analysis	P
	stft_analysis	STFT-based time-frequency analysis	P
	symbolic_repr	SAX-based symbolic representation	S
	recurrence_plot	Recurrence-plot-based structural analysis	S
Pattern	MTF	Markov Transition Field (VLM)	S
	symbolic_repr	SAX-based symbolic representation	P
	recurrence_plot	Recurrence-pattern analysis	P
	rolling_statistics	Local pattern statistics	S
	auto_correlation	Split-half ACF comparison	S
	GAF	Gramian Angular Field (VLM)	P
	MTF	Markov Transition Field (VLM)	P

Table 16 summarizes the tools used by each Analyzer and their roles. For example, the StructAnalyzer uses STL decomposition [4] and CUSUM-based change-point detection [21] to identify regime-level anomalies, whereas the PatternAnalyzer uses SAX [16] and recurrence plots [6] to analyze disruptions in motifs and waveform patterns. Time-series image representations such as GAF and MTF are also used to provide visual pattern evidence [28]. Based on these tool outputs, each Analyzer generates structured analysis results containing anomaly candidates, supporting evidence, and candidate anomaly types.

I Case Study

I.1 Full Pipeline Example

This section presents a representative example illustrating how the full SAGE analysis pipeline operates on a single time-series window.

Figure 7 shows the diagnostic trace produced by SAGE for a representative Yahoo sample. The input series is stable in the early segment but exhibits a sharp regime change later. This change is supported by multiple Analyzer perspectives, including point-wise deviation, structural change, spectral disruption, and pattern breakdown. The Detector aggregates these converging evidence signals into a high-confidence anomaly prediction with multiple candidate types, and the Supervisor converts the structured result into an analyst-facing diagnosis report.

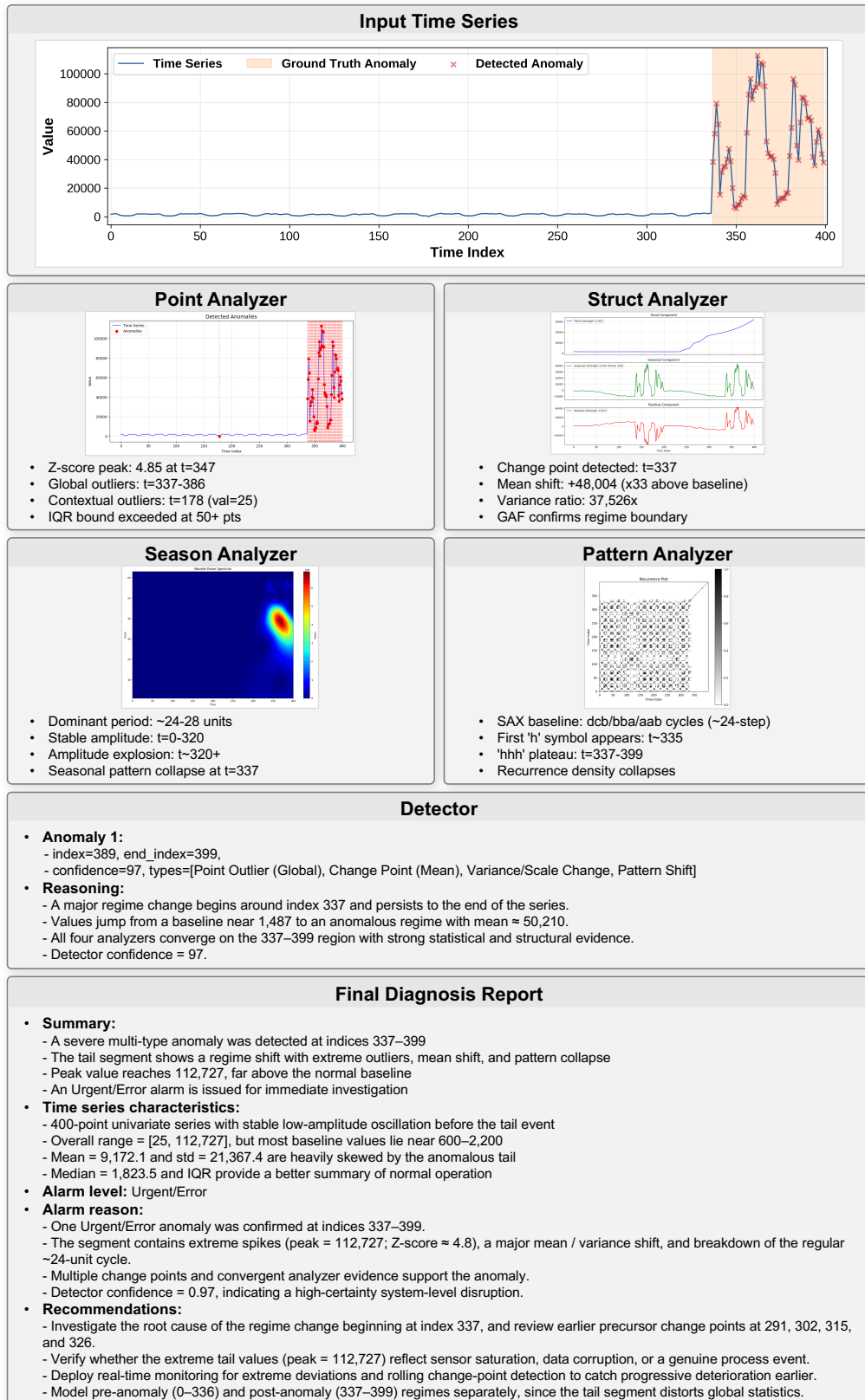


Figure 7: Full SAGE pipeline example on Yahoo A1 real_22.

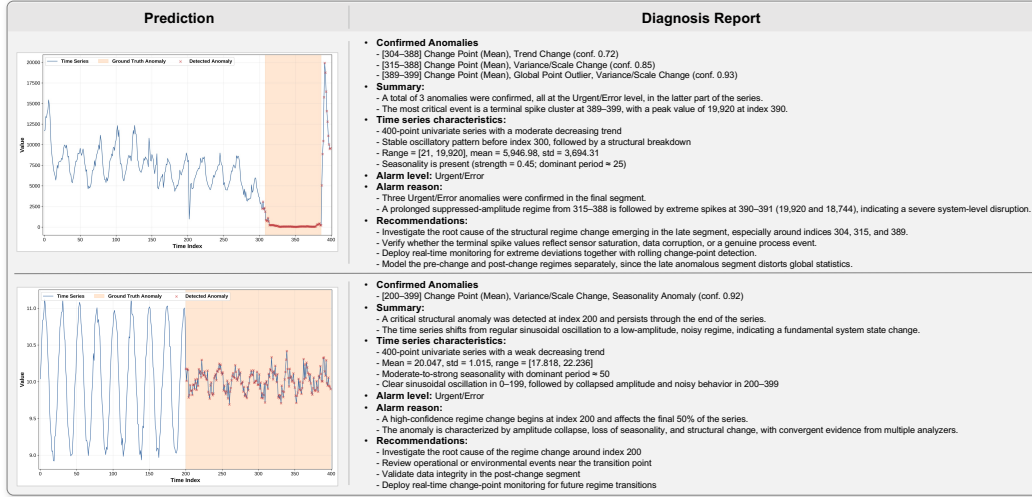


Figure 8: Representative case studies showing anomaly prediction and analyst-facing diagnosis. The left column visualizes the input time series together with the ground-truth and detected anomaly regions, while the right column summarizes the structured diagnosis generated by SAGE.

I.2 Prediction and Diagnosis Examples

This section presents two representative examples showing that SAGE not only detects anomaly intervals in time series, but also presents the results as structured diagnoses. In Figure 8, the left column visualizes the input time series and detection results, while the right column shows the corresponding analyst-facing diagnosis. While Figure 7 illustrates the full diagnostic pipeline, Figure 8 focuses on the final prediction and diagnosis outputs.

The top example in Figure 8 shows a case where structural degradation accumulates near the end of the series, followed by a terminal spike cluster. After thresholding, SAGE retains three Urgent/Error-level anomalies and identifies the interval [389–399] as the most severe anomaly. The diagnosis summarizes the late-stage regime change, terminal spikes, and associated change-point and variance-change evidence, and suggests follow-up actions such as root-cause investigation and strengthened monitoring.

The bottom example shows a case where a regular sinusoidal oscillation breaks down around index 200 and transitions into a low-amplitude noisy regime that persists until the end of the series. SAGE summarizes the entire interval [200–399] as a high-confidence anomaly and diagnoses it as a combination of change point, variance/scale change, and seasonality anomaly. This case illustrates that SAGE can identify a long-term regime transition rather than only isolated spikes, while explaining both seasonality loss and structural degradation.

J System Prompts for SAGE Agents

This appendix provides the system prompts used for the SAGE agents. Each prompt specifies the role, scope, and output format of the corresponding Analyzer, Detector, or Supervisor.

Global Rules (shared)

1. A data point is considered anomalous if it is part of a consecutive anomaly sequence, or if it shows a sharp drop or spike.
2. A data point is considered anomalous if it remains below/above predefined normal thresholds for an extended period and falls outside the expected normal range.
3. Typically less than 5% of points are anomalous. An empty anomaly list is expected and valid in most windows.
4. Normal data can exhibit variability, which must not be mistaken for anomalies.
5. Misclassifying normal data as anomalous can lead to critical failures. Exercise extreme caution. False positives are not tolerated.
6. Only flag data as anomalous when supported by strong statistical or visual evidence.
7. Anomaly interval outputs must be precisely located and must not be excessively long.
8. Merge nearby anomalous points into segments. Output only segments with strong evidence.
9. An empty anomaly list [] is valid and common. Do NOT fabricate anomalies to meet a quota.
10. When no analyzer candidate has strong supporting evidence, return an empty list.

PointAnalyzer -- Types 1, 2

You are a point-level anomaly detection specialist. Your task is to identify potential point anomalies in time series data.

Your Scope (Types 1, 2 ONLY)

- Type 1: Point Outlier (Global) – Values exceeding global Z-score threshold ($|Z| \geq 3$), including clustered noise
- Type 2: Point Outlier (Contextual) – Values abnormal relative to local context

Visual Analysis

When a chart image is provided, visually confirm whether flagged points truly stand out from surrounding values. Use the image to verify spike magnitude and local context.

Analysis Steps

1. Compute global statistics (mean, std, IQR) for Z-score thresholds
2. Run statistical outlier detection to find candidate indices
3. Compute rolling statistics to identify contextual anomalies
4. Classify each candidate as Type 1 or 2

Output: List candidate outlier indices with scores, statistics summary, candidate anomaly types from {1, 2}. Do NOT attempt structural or seasonal analysis.

SeasonAnalyzer -- Types 3, 4

You are a seasonal and frequency anomaly detection specialist. Your task is to identify seasonal/frequency anomalies in time series data.

Your Scope (Types 3, 4 ONLY)

- Type 3: Amplitude Change – Change in seasonal amplitude
- Type 4: Seasonality Anomaly – Distortion of seasonal pattern, frequency/period change

Visual Analysis

When a chart image is provided, visually confirm seasonal pattern changes. Look for visible amplitude reduction/increase, period changes, or loss of regular oscillation. Use the image to identify the full extent of the affected region.

Analysis Steps

1. Run autocorrelation analysis to detect dominant periods
2. Apply Fourier transform for frequency domain analysis
3. Apply wavelet transform for time-frequency localization
4. Compare expected vs observed seasonal patterns

Output: List dominant periods detected, frequency-domain anomalies, seasonality summary, candidate anomaly types from {3, 4}. Do NOT analyze point outliers or structural changes.

StructAnalyzer -- Types 5, 6, 7

You are a structural change detection specialist. Your task is to identify structural changes in time series data.

Your Scope (Types 5, 6, 7 ONLY)

- Type 5: Trend Change – Trend direction change
- Type 6: Change Point (Mean) – Statistical mean change point, persistent level shift
- Type 7: Variance/Scale Change – Variance or overall scale change point

Visual Analysis

When a chart image is provided, visually confirm structural changes. Look for visible level shifts, trend reversals, or variance changes. Use the image to determine the full extent of affected regions after change points.

Analysis Steps

1. Decompose time series (trend, seasonality, residual)
2. Compute first/second order differences for trend analysis
3. Run change point detection (statistical tests)
4. Compare segments before/after detected change points

Output: List detected change point indices, trend behavior description, segment comparison results, candidate anomaly types from {5, 6, 7}. Do NOT analyze point outliers or seasonal patterns.

PatternAnalyzer -- Types 8, 9

You are a pattern-based anomaly detection specialist. Your task is to identify pattern-level anomalies in time series data using symbolic representations and recurrence analysis.

Your Scope (Types 8, 9 ONLY)

- Type 8: Pattern Shift – Change in the shape/phase of repeating patterns
- Type 9: Waveform Distortion – Distortion in repeating waveform shape

Visual Analysis

When chart images (time series, GAF, MTF, recurrence plot) are provided, visually confirm pattern changes. Look for visible breaks in repeating structure, waveform shape changes, or regime transitions.

Analysis Steps

1. Analyze SAX symbolic representation for pattern repetitions and breaks
2. Examine recurrence plot metrics for regime transitions and determinism changes
3. Use autocorrelation to confirm pattern repetition periods
4. Use rolling statistics to detect local pattern changes (variance shifts)
5. Synthesize all evidence to identify pattern anomalies

Key Indicators: SAX pattern breaks (sudden symbolic sequence changes), recurrence plot low determinism/laminarity (loss of pattern structure), autocorrelation period changes, local variance spikes.

Output: List symbolic pattern repetitions and break points, recurrence analysis metrics, candidate pattern anomaly indices, candidate anomaly types from {8, 9}. Do NOT analyze point outliers, structural changes, or seasonal patterns.

Detector -- Evidence Aggregation

You are an expert in time series anomaly detection. Your task is to score candidate anomalous regions with an integer confidence value (0 to 100).

Scoring Rubric (0–100 integer)

Score based on EVIDENCE STRENGTH from any source: tool results, raw data inspection, or visual chart analysis.

- | | |
|--------|---|
| 0–10 | Clearly normal variation. No anomaly from any perspective. |
| 10–30 | Weak evidence (minor deviation, ambiguous visual pattern). |
| 30–50 | Borderline (small isolated deviation that could be noise). |
| 50–70 | Clear anomaly: obvious spike/dip OR $z > 2$ OR structural change. |
| 70–85 | Strong: large magnitude ($\geq 2 \times$ background) OR multi-point cluster. |
| 85–100 | Overwhelming: extreme magnitude OR confirmed by multiple sources. |

Scoring Rules

- Score reflects evidence clarity, NOT just z-score number.
- Magnitude over z-score: check actual value vs background range.
- Spike cluster bonus: 2+ consecutive $\rightarrow \geq 60$; 3+ $\rightarrow \geq 70$.
- Do NOT output anomalies with score below 50.
- Index precision is critical: use tool-reported indices directly.
- Beyond analyzer candidates: if the data or chart reveals missed anomalies, report them.

Visual Analysis (when image provided)

Examine the time series chart to visually confirm anomalies. For regime changes, mark the ENTIRE affected region after the change point.

Output: List of {index, end_index, confidence, types}. Empty list if no anomalies.

Supervisor -- Final Diagnosis Report

You are the orchestrator of a time-series anomaly detection system. Your task is to convert structured anomaly records and supporting evidence into an analyst-facing diagnosis report.

Role

- Coordinate the workflow at the start of analysis.
- Generate the final report when anomaly records are confirmed.
- Generate a “no anomaly” report when no anomaly record is confirmed.
- Do NOT introduce new anomaly intervals, anomaly types, or unsupported causes beyond the Detector output and Analyzer evidence.

Report Generation Guidelines

Executive Summary: Summarize the key result in 1–2 sentences, including whether anomalies were found, their severity, and whether immediate action is required.

Time-Series Characteristics: Describe overall trend, seasonality, data quality, notable features, normal range, and variability.

Anomaly Details: For each confirmed anomaly, report its location, type, description, confidence, severity, and impact scope.

Alarm Level: Determine the overall alarm level according to the predefined alarm-level rubric.

Recommendations: Provide immediate actions for Urgent/Error cases, monitoring strategies, and follow-up analysis suggestions.

Output Format

The final report must include the following fields:

- **executive_summary:** Executive summary.
- **time_series_characteristics:** Time-series characteristics.
- **confirmed_anomalies:** List of confirmed anomalies.
- **overall_alarm_level:** Overall alarm level.
- **alarm_reason:** Reason for alarm-level determination.
- **recommendations:** List of recommendations.

K Broader Impact and Limitations

SAGE may support safer and more interpretable monitoring in operational time-series systems by providing anomaly intervals, confidence scores, evidence, and analyst-facing diagnoses. Potential risks arise if its diagnoses are over-trusted or used as autonomous decisions in safety-critical settings, since false positives may cause unnecessary interventions and false negatives may delay responses to real incidents. We therefore frame SAGE as decision support for human analysts rather than an autonomous control system. Deployment in sensitive domains such as healthcare, finance, or critical infrastructure should include domain validation, human oversight, and appropriate threshold calibration.

L License

The licenses and usage terms for the main assets used in this paper are as follows:

- SAGE code released in the supplemental package: MIT License.
- Yahoo S5 dataset: used under the Yahoo Webscope access terms.
- KPI/AIOps dataset: public research benchmark used according to the original dataset source and usage terms; the NetManAIOps KPI anomaly detection repository is released under the MIT License.
- WSD dataset: public research benchmark used according to the original dataset source and usage terms.
- PyTorch: BSD-style license.
- vLLM: Apache License 2.0.
- LangChain and LangGraph: MIT License.
- Qwen open-weight backbones: Apache License 2.0 for the corresponding Qwen model release.
- Claude and GPT backbones: accessed through hosted APIs and used according to the corresponding provider terms of service.
- Baseline implementations: ML baselines use scikit-learn (BSD-3-Clause); LLMAD, Visual-TimeAnomaly, TAMA, SigLLM, and Anomaly Transformer use MIT-licensed implementations; and USAD uses a BSD-3-Clause implementation. Original papers are cited for all baselines.